

Executive Summary

Event Horizon now has a first playable vertical slice on feat/first-playable: a mobile-first Vite + PixiJS v8 canvas game where the player taps and swipes to capture dark-energy orbs, delay collapse, export a share poster, and submit replay-shaped scores.

Assumptions

- Greenfield repo; plain Vite was the simplest viable scaffold.
- Seed date remains eh-2026-05-29-alpha from the task context.
- GitHub Pages uses /event-horizon/; Netlify uses VITE_BASE_PATH=/ via netlify.toml.
- Custom deterministic motion is enough; Matter.js or Planck.js is not needed for this slice.
- Score endpoint validates only; persistence and anti-cheat are backlog work.

Chosen Stack and Why

- Vite + TypeScript for a small static site and typed modules.
- PixiJS v8 for WebGL-preferred 2D rendering with generated textures and a scene graph.
- Fixed 60 Hz custom simulation for replayable seed + input-timing runs.
- Vitest and Playwright Chrome for deterministic, endpoint, touch, poster, and smoke checks.
- Netlify function plus GAS sample for minimal score endpoint options.

Plan and Estimates

Repo guide 0.25h; scaffold 0.75h; simulation/replay 2h; Pixi scene/HUD 2h; input 1h; collapse/poster 1.25h; endpoints 0.75h; tests/artifacts 1.25h; docs/report 1.25h. Total: 10.5h.

Backlog

- Durable score storage and abuse limits.
- Replay playback UI and CI replay verifier.
- Balance phase thresholds, spawn curves, scoring, sound, haptics, and reduced motion.
- GitHub Actions for build, lint, unit, and Chrome smoke tests.

Files Created or Changed

AGENTS.md, README.md, index.html, package.json, package-lock.json, tsconfig.json, vite.config.ts, eslint.config.js, playwright.config.ts, netlify.toml, docs/*, gas/score-submit.gs, netlify/functions/score-submit.mjs, scripts/*, src/main.ts, src/styles.css, src/vite-env.d.ts, src/game/*, tests/.*

Key File Notes

- Simulation.ts: deterministic state, orb/flyby/shadow-arm mechanics, replay payloads.
- EventHorizonGame.ts: PixiJS scene, scaling, HUD, collapse, poster export, score submit.
- FixedStepLoop.ts and rng.ts: 60 Hz decoupled updates and mulberry32 RNG.
- InputHandler.ts: pointer tap/swipe mapping to 1080 x 1920 logical space.
- posterizer.ts: vertical share image from three gameplay frames.
- score-submit.mjs and score-submit.gs: Netlify and GAS score endpoints.
- tests/.*: deterministic replay, endpoint, mobile Chrome, touch, poster, and collapse checks.

Sample Replay Payload

```
{"version":1,"seed":"eh-2026-05-29-alpha","startedAt":1780051200000,"survivalMs":67421,"score":1280,"energyCaptured":87,"maxStreak":24,"tapEvents":[],"swipeEvents":[],"phaseTransitions":[]}
```

Test Results

- npm run build: passed; Vite dist generated.
- npm run lint: passed.
- npm run test: passed, 2 files and 5 tests.
- npm run test:e2e: passed, 4 mobile Chrome tests.
- npm run score:test: passed, status 200.
- Local Chrome smoke, deterministic replay, touch simulation, poster export, collapse, and score submit all passed.

Performance Notes

Stable art uses generated textures; background is static; hot-loop allocations are minimized; no expensive filters or physics engine. Poster capture uses toDataURL only on user action.

Run, Git, and Deployment

```
npm install
npm run dev
npm run build
npm run lint
npm run test
npm run test:e2e
npm run score:test
npm run capture:artifacts
npm run report:pdf
git switch -c feat/first-playable
git add .
git commit -m "Build first playable Event Horizon slice"
git push -u origin feat/first-playable
gh pr create --base main --head feat/first-playable --title "Build first playable Event Horizon slice" --body-file docs/pr-body.md
```

Netlify: build command `VITE_BASE_PATH=/ npm run build`, publish dist, functions directory `netlify/functions`, endpoint `/.netlify/functions/score-submit`, use `npm run dev` locally. GitHub Pages: default Vite base is `/event-horizon/` for `https://kevinhegg.github.io/event-horizon/`.

Major Snippets

```
await app.init({ autoStart:false, preference:"webgl", resizeTo:this.root, preserveDrawingBuffer:true });
while (accumulatorMs >= FIXED_STEP_MS) step(FIXED_STEP_MS);
export function mulberry32(seed){ state=(state+0x6d2b79f5)>>>0; /* deterministic */ }
if (moved >= swipeThresholdSq) onSwipe(start,end); else onTap(end);
function doPost(e){ return ContentService.createTextOutput(JSON.stringify({ok:true})).setMimeType(ContentService.MimeType.JSON); }
```

PR Summary

PR #1 is opened from `feat/first-playable` to `main` with project scaffold, PixiJS canvas integration, deterministic loop, gameplay slice, endpoints, tests, screenshots, README, AGENTS.md, and this PDF.

Screenshots and Poster

Gameplay



Poster



Collapse



Diff Stat

```
docs/artifacts/collapse-mobile.jpg | Bin 5805 -> 2601 bytes
docs/artifacts/gameplay-mobile.jpg | Bin 4725 -> 2119 bytes
docs/artifacts/share-poster.jpg    | Bin 6992 -> 2815 bytes
scripts/capture-artifacts.mjs      | 12 +-
scripts/generate-report-pdf.mjs    | 248 ++++++-----
5 files changed, 131 insertions(+), 129 deletions(-)
```

Diff Excerpt

[Excerpt capped for PDF practicality; full diff is in PR #1.]

